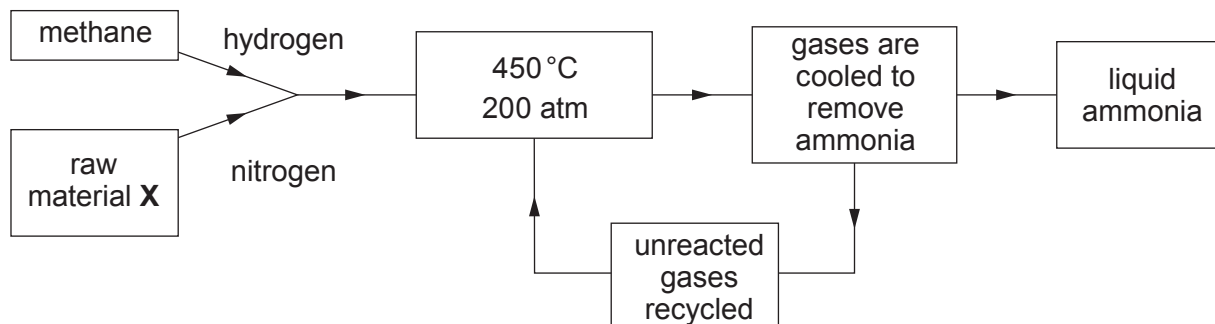


9. (a) The diagram outlines the manufacture of ammonia by the Haber process.



- (i) Name the raw material **X**. [1]
- (ii) The pressure used in the Haber process is 200 atm. State why a higher pressure is **not** used. [1]
-
- (iii) At 450 °C, the reaction is very slow. Iron is used in the process to speed up the reaction. Give the name for a substance used to speed up a chemical reaction. [1]
-
- (iv) The reaction between nitrogen and hydrogen is represented by the equation below.



Complete the equation below using the key:

[2]



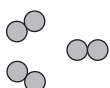
nitrogen gas, N_2



hydrogen gas, H_2



+



- (b) One of the main uses of ammonia is in the manufacture of fertilisers.

The table shows the results obtained when tests were carried out on three different fertilisers **A**, **B** and **C**.

Fertiliser	Test for positive ion	Test for negative ion
A	On adding sodium hydroxide solution and warming, a pungent smelling gas is formed which turns red litmus blue	On adding barium chloride solution a white precipitate forms
B	Lilac flame test	On adding silver nitrate solution a white precipitate forms
C	On adding sodium hydroxide solution and warming, a pungent smelling gas is formed which turns red litmus blue	On adding silver nitrate solution a white precipitate forms

Give the **letter** of the fertiliser which is ammonium sulfate.

[1]

Letter

- (c) Ammonia reacts with chlorine to form nitrogen and hydrogen chloride.

Complete the balancing of the equation for this reaction.

[1]

